How Much Energy Does a Modal Contact Row Inject? A Cross-Formulation Measurement and a Cumulative Storage Bound

Anonymous Author(s)

Abstract

Reduced (modal) deformable bodies resting in unilateral contact with rigid bodies are usually resolved by a velocity-level complementarity law that couples rigid and modal unknowns through one shared multiplier. Interactive solvers run that law at small, fixed local iteration budgets. We measure what this costs. Running one transcribed contact row through three fixed-budget solver formulations — position-based (XPBD), augmented-Lagrangian (AVBD), and an implicit sequential-impulse realization — over an identical 24-cell scene × relaxation × budget sweep, we find sharply formulation-dependent behaviour. Against the measured gross rigid-side loss (scene-wide, gravity-corrected, and not a per-contact dissipation), the position-based host overdraws by up to 4.4×10^7 J — 1.2×10^5 times the incident rigid kinetic energy in the most starved cells — the augmented-Lagrangian host overdraws in 23 of 24 cells but by at most 15 J, and the implicit realization never overdraws at all. Pervasive but negligible and rare but catastrophic are different pathologies, and the customary ratio-to-incident-energy diagnostic does not separate them. A budget sweep shows the amplification is a truncation artifact: it decays monotonically toward a converged reference computed in the same code path, and the gap to that reference shrinks by six orders of magnitude. Because the converged regime is unaffordable at interactive budgets, we add a cumulative modal-storage bound funded by that measured gross rigid-side loss, enforced by a reservoir ledger and a projection of the realized modal state, which holds in all 72 measured cells (60 distinct configurations). We also measure what that projection costs contact validity, and find worst-case post-projection penetration of 21.6 mm — 72% of the supporting board's thickness — with corrective impulses up to 8.7× steady state.

CCS Concepts

• **Computing methodologies → Physical simulation**.

Keywords

contact, modal reduction, passivity, position-based dynamics

ACM Reference Format:
Anonymous Author(s). 2026. How Much Energy Does a Modal Contact Row Inject? A Cross-Formulation Measurement and a Cumulative Storage Bound. In *The 19th ACM SIGGRAPH Conference on Motion, Interaction and Games (MIG '26), December 11–13, 2026, Charleston, SC, USA.* ACM, New York, NY, USA, 7 pages.

1 Introduction

A deformable object that is struck at one point and rings elsewhere is routinely modelled by projecting the elastic response onto a small modal basis. When such a body also rests in unilateral contact with rigid bodies, the established treatment resolves rigid and modal unknowns together through a shared contact multiplier at the velocity level [9]. The one-way alternative — driving a separate modal filter from contact events without back-reaction — is the real-time precedent [1, 5].

We take that coupled row as given. Written at position level it is one linearization step from the velocity-level law: for gap function C, $C^{n+1} \approx C^n + h\mathbf{J}\mathbf{u}^{n+1}$, which divided by h is the velocity-level complementarity condition with a C^n/h stabilization bias, carrying the same Jacobian and the same multiplier as Zheng and James [9]. *The row is not our contribution*, and ours is narrower than theirs: modes couple through normal rows only, and the position-level form takes $e = 0$ restitution.

Interactive solvers do not solve that row to convergence. They run a fixed, small number of local iterations per substep, and it is known that finite-iteration coupling across a partition boundary can inject spurious energy even when every subsystem is individually passive [7]. *That observation is not ours either.* What is missing is a quantitative answer for this specific case — unilateral contact-to-modal transfer — and a mitigation that fits inside an interactive budget.

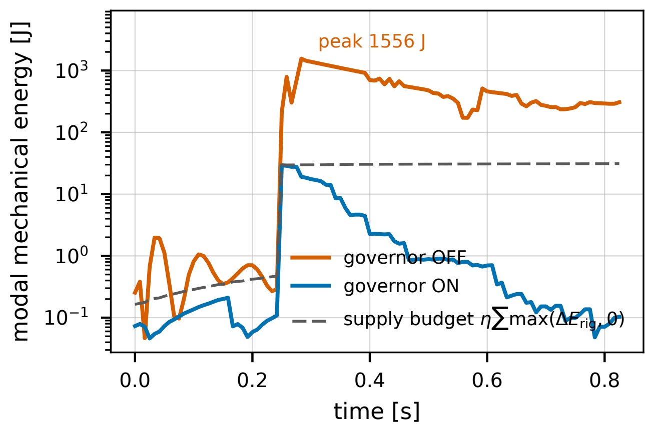

Figure 1: The same impact at the same starved budget (8×2), with and without the cumulative storage bound; the traces are identical until contact. Un-governed, modal energy runs to 1556 J in a scene whose incident rigid kinetic energy is ∼29 J; governed, it peaks at 29.3 J and stays under the running supply budget that funds it (dashed), which is measured gross rigid-side loss, not a tuned constant.

Contributions.

(1) **Measurement.** One contact row, three fixed-budget solver formulations, one identical 24-cell adversarial sweep, one machine (§3.1); plus an iteration-budget sweep against a converged reference showing the injection is a truncation artifact (§3.2).
(2) **A cumulative storage bound.** A one-directional bound on energy entering modal storage, funded by measured gross rigid-side loss and enforced by a reservoir ledger and a projection of the realized modal state (§2); it satisfies the prescribed inequality in all 72 measured cells (60 distinct configurations).
(3) **The cost of enforcement.** Post-projection contact validity — penetration, corrective impulse, multiplier variance, momentum — measured rather than assumed (§3.3).

Relation to concurrent energy-control work. Wei et al. [7] guarantee discrete passivity for any finite inner-iteration budget by construction, via Douglas–Rachford splitting in wave coordinates whose Fejér monotonicity supplies algorithmic dissipation; their interfaces are *bilateral* couplings, not unilateral contact, and the guarantee is structural rather than a budgeted transfer cap. You et al. [8] modify implicit Euler to track user-specified *total*-energy targets bidirectionally on full nonlinear elastodynamics with barrier contact. Ours is neither: a one-directional cap on what contact may deposit into modal storage, funded by measured gross rigid-side loss, enforced by scaling realized modal state. For flexible bodies in a game engine, Zobel et al. [10] solve ray-traced penalty contact quasi-statically without a shared unilateral multiplier; modes and modal derivatives have been used in a position-based host as a *deformation* surrogate [6], whereas here the modal amplitudes enter as contact DOFs driven by the shared multiplier.

Relation to passivity control and energy projection. The governor transplants an established control idea rather than inventing one. Time-domain passivity control [4] damps once an energy observer goes negative; energy-tank formulations [3] give that observer explicit state, so a controller spends only what its tank holds. We keep that structure and change two things: the tank is funded by *measured gross rigid-side loss* — energy the solve removed elsewhere — rather than by the port it regulates, and the actuator scales realized *state* rather than modulating a force, because a fixed-budget solver commits its multipliers before the excess is observable. FEPR [2] also projects state post-solve, but onto conservation of *total* energy; ours is one-sided and never restores energy the solver lost.

2 The row and the cumulative bound

Let $\mathbf{q} \in \mathbb{R}^r$ be mass-normalized modal amplitudes of the supporting body, so its modal mechanical energy is $E_{\text{mod}} = \frac{1}{2}\dot{\mathbf{q}}^\top \dot{\mathbf{q}} + \frac{1}{2}\mathbf{q}^\top \mathbf{K}\mathbf{q}$. A support-contact row between a rigid corner at world height y_c and the deformed surface has gap

$$C \;=\; y_c \;-\; \left(y_{\text{rest}} + \mathbf{U}_y^\top \mathbf{q}\right), \qquad C \geq 0,\ \lambda \geq 0,\ C\lambda = 0, \tag{1}$$

with $\partial C/\partial \mathbf{q} = -\mathbf{U}_y$, so one multiplier λ drives both the rigid body and the modal amplitudes. Equation (1) is the transcription discussed above, not a new law.

The invariant. We require that the energy *stored in the modal subsystem* never exceed the *measured gross rigid-side loss* — defined

1 snapshot E^-_{rig} (pure KE, all bodies), E^-_{mod}, positions $\mathbf{x}^-$
2 predict (gravity); generate contacts at the predicted pose
3 position solve (K iterations); velocity solve *the contact solve*
4 E^+_{rig}; $\Delta E_{\text{rig}} = (E^-_{\text{rig}} - E^+_{\text{rig}}) + W_g$ (3), *post-solve*
5 $B \leftarrow B + \eta \max(\Delta E_{\text{rig}}, 0)$ *credit, same substep*
6 E^+_{mod}; **if** $E^+_{\text{mod}} > E^-_{\text{mod}} + B$:
7 $\quad \gamma = \sqrt{(E^-_{\text{mod}} + B)/E^+_{\text{mod}}}$; $(\mathbf{q}, \dot{\mathbf{q}}) \leftarrow \gamma(\mathbf{q}, \dot{\mathbf{q}})$ (4)
8 $B \leftarrow \max(B - \max(\Delta E_{\text{mod}}, 0),\ 0)$ *debit realized gain*

Contact is **not** re-solved after step 7 — the projection moves the surface of (1) and the substep ends (§3.3 measures the cost).

Figure 2: One substep of the governed loop, identical in all three hosts. The reservoir is credited *after* the contact solve, from the loss that solve produced.

exactly at (3) below — cumulatively over the run:

$$E^n_{\text{mod}} - E^0_{\text{mod}} \;\;\leq\;\; \eta \sum_{k \leq n} \max\!\left(\Delta E^k_{\text{rig}},\ 0\right), \qquad \eta \in (0, 1], \tag{2}$$

The transfer efficiency η sets how much of the measured loss may be spent; *every experiment in this paper runs* $\eta = 1$, the least restrictive admissible value, so no result below depends on tuning it. The bound is *cumulative*, not per-step: a substep may deposit more than it earns provided the running total stays inside the budget, which is what lets a legitimate ring persist across substeps.

Every term, exactly. Modal storage is the mechanical energy of the reduced state, $E_{\text{mod}} = \frac{1}{2}\dot{\mathbf{q}}^\top \mathbf{M}_q \dot{\mathbf{q}} + \frac{1}{2}\mathbf{q}^\top \mathbf{K}_q \mathbf{q}$, and $E^0_{\text{mod}} = 0$: every scene starts from an undeformed, resting support, so the run's own settling transient must be funded like anything else. The supply is

$$\Delta E^k_{\text{rig}} \;=\; \left(E^{k,-}_{\text{rig}} - E^{k,+}_{\text{rig}}\right) + W_g{}^k, \qquad W_g{}^k = \sum_b m_b\, \mathbf{g}\left(\mathbf{x}^{k,+}_b - \mathbf{x}^{k,-}_b\right), \tag{3}$$

where $E_{\text{rig}} = \sum_b \left[\frac{1}{2} m_b \|\mathbf{v}_b\|^2 + \frac{1}{2}\boldsymbol{\omega}_b^\top \mathbf{I}_b \boldsymbol{\omega}_b\right]$ is *purely kinetic* — translational and rotational, over all dynamic bodies, not merely those in contact — and $\pm$ denote the two ends of substep k. We call this the **measured gross rigid-side loss**: the scene-wide, gravity-corrected, gross kinetic-energy loss of the rigid subsystem. It is a *loss*, not a dissipation — it also counts legitimate rigid-to-modal transfer and losses at rigid–rigid contacts elsewhere in the scene — so it is an envelope on what contact dissipated, not a measurement of it (§4). Gravity enters only through the work term W_g, which makes free ballistic motion register zero supply: a falling body's kinetic gain exactly equals the gravity work, so nothing is credited and only an inelastic velocity change funds the reservoir. Endpoints are captured at substep boundaries and tile the timeline exactly, so no rigid energy change escapes the accounting.

What is guaranteed, and what is not. Equation (2) is a *cumulative, gross-loss-funded storage ceiling on the modal subsystem*, with a supply that is *measured, not prescribed*: the right-hand side is the measured gross rigid-side loss of (3), never a tuned constant or a fraction of incident energy. It is not contact-port passivity and not a signed per-interface transfer bound — it constrains a scalar reservoir, not a port. The *gross* sum $\sum \max(\Delta E_{\text{rig}}, 0)$ has two consequences worth stating: it ignores substeps in which the rigid subsystem *gains* energy, so those cannot cancel earlier credits, and energy returned to the rigid bodies and dissipated again counts as fresh supply. We measure the second effect in §4.

The enforced inequality is slightly weaker than the stated one. Reported violations use (2) exactly as printed. The implementation's own test additionally forgives one substep's largest deposit, $\max_k \eta \max(\Delta E^k_{\mathrm{rig}}, 0)$, absorbing the same-substep granularity artifact of Fig. 2: modal potential energy can rise in the very substep whose credit funds it. In these scenes that allowance is worth 7–389 J — immaterial where the overdraw is 10^5 J or more, decisive where it is not: under it the augmented-Lagrangian host would be recorded as violating 1 of 24 cells rather than 23. We report the stricter reading throughout, for both governed and un-governed columns, so that one inequality adjudicates both; the governed runs satisfy it with a worst margin of 1.1×10^{-13} J.

Enforcement. A reservoir B accumulates the supply and is debited by realized storage. Both happen *after* the contact solve, in the same substep (Fig. 2): ΔE_{rig} is not observable until the velocity solve has finalized the rigid velocities, so the substep credits $B \leftarrow B + \eta \max(\Delta E_{\mathrm{rig}}, 0)$ at that point, tests the realized modal energy against it, and only then debits $B \leftarrow \max(B - \max(\Delta E_{\mathrm{mod}}, 0), 0)$. Crediting within the funding substep is precisely what lets modal potential energy rise in the substep that pays for it — the granularity artifact the forgiveness above absorbs. If the realized modal energy would overdraw B, the realized state is projected onto the admissible set by a single scalar,

$$(\mathbf{q}, \dot{\mathbf{q}}) \;\leftarrow\; \gamma\,(\mathbf{q}, \dot{\mathbf{q}}), \qquad \gamma \;=\; \min\!\left(1,\; \sqrt{\frac{E^-_{\mathrm{mod}} + B}{E^+_{\mathrm{mod}}}}\right), \tag{4}$$

with $E^\mp_{\mathrm{mod}}$ the modal energy before and after the solve. The form is closed because E_{mod} is quadratic in the state: scaling $\mathbf{q}$ and $\dot{\mathbf{q}}$ together scales both the kinetic and the elastic term by γ^2, so γ bounds the total rather than only the velocity. Two consequences matter and are measured below: the projection is inert whenever the solve is already within budget ($\gamma = 1$, bitwise-identical trajectories), and it does not re-solve contact (Fig. 2).

3 Results

Every solver-behaviour number below was produced on one machine (Table 1); the sole exception is the device-resident timing paragraph of §3.5 (NVIDIA RTX 4090), reported separately. All are frozen with generating commands and commit hashes in the supplemental ledger. Cross-machine mixing is deliberately avoided: chaotic contact stacks diverge between architectures under floating-point reassociation.

3.1 One row, three formulations, the same 24 cells

We drive an impactor into a modally-reduced support in three scenes — a shelf board, a ledge slab, and a 2.2 × 1.1 m table, the last two following the layouts of Coevoet et al. [1] — and sweep modal relaxation and the local budget (iterations × substeps) over {4×1, 8×2, 16×4, 32×8}. Table 1 lists what the hosts do differently and what is pinned identical. Two entries deserve emphasis, being the ones a controlled comparison could be accused of getting wrong. The relaxation axis is *inert* on the impulse host — its implicit modal weight needs no under-relaxation and the attribute is never read — so its two relaxation rows are bit-identical by construction, not coincidence. And warm start differs, which is intrinsic to the formulation rather than a knob we chose: the position-based host zeroes its multipliers every substep, the other two carry them across.

Table 1: What differs between the three hosts, and what does not. Everything below the rule is held identical; everything above it is intrinsic to the formulation. Entries are read from the implementation, with source anchors in the supplemental ledger.

	position-based	augmented-Lagr.	seq. impulse
unknown solved for	position	position + multiplier	velocity
contact treatment	compliant projection, $\tilde{\alpha} = \alpha/h^2$ (α=0: rigid)	penalty + dual update, α=0.99, β=10^5	Schur system, $\mathbf{A} = \frac{\mathrm{cfm}}{h^2}\mathbf{I} + \mathbf{J}\mathbf{M}^{-1}\mathbf{J}^\top + \hat{\mathbf{G}}\mathbf{W}\hat{\mathbf{G}}^\top$
modal weight	explicit; per-mode compliance $1/(H_{ii}h^2 - 1)$	block-GS, full Newton step on $\mathbf{q}$	*implicit*, $\mathbf{W} = (\mathbf{M} + h\mathbf{D} + h^2\mathbf{K})^{-1}$
iteration	GS sweeps, gap re-evaluated each sweep	coloured primal GS + $\mathbf{q}$-block	PGS over rows
warm start	none (λ←0 each substep)	λ and penalty carried over	λ cache keyed per row
relaxation	0.7 on $\mathbf{q}$ and the support block	0.7 on the $\mathbf{q}$-block Newton step	*inert*: never read
h; substep	1/120 s; h/S, contacts regenerated per substep		
modal rank r	24 (shelf), 28 (ledge), 24 (table)		
modal damping	Rayleigh $\mathbf{D} = \alpha_0\mathbf{M} + \alpha_1\mathbf{K}$, $\alpha_0 \in \{2, 3\}$, α_1=10^{-5}		
stepper; η	implicit midpoint; $\eta = 1$		
machine	Apple M4, CPU only, CPython 3.12		

Equal work, unequal outcome. A cell costs $K \cdot S$ row evaluations per frame, so the budget axis quadruples the work at every rung (4, 16, 64, 256) and spans ×64 — steeper than its labelling suggests — and it conflates two refinements that are not interchangeable. At a fixed 32 row evaluations on the shelf drop, spent as iterations (K=32, S=1) the ratio is 0.300 and (2) holds; spent as substeps (K=4, S=8) it is 3.13, overdrawing by 481 J. Substep refinement alone does not converge the contact row (full ladder in the supplement).

Figure 3 shows the outcome. The position-based host exceeds $R = 1$ in 8 of 24 cells, worst case 1.19534×10^5 (ledge, relaxation 1.0, 4×1: peak modal energy 4.44×10^7 J against an incident 371 J). The augmented-Lagrangian host exceeds $R = 1$ in 2 of 24 — both at the starved 4×1 corner of the table scene, worst 1.70 — so it is *not* unconditionally benign either. The ordering is scene-dependent and it inverts: worst-over-scenes at 4×1 the position-based host is five orders of magnitude more energetic (1.2×10^5 vs 1.70), but *on the table scene itself* the augmented-Lagrangian host is the worse (1.18 and 1.70 against a position-based worst of 0.37). Neither host dominates across scenes.

The ratio is not the invariant. R is a severity diagnostic: one impactor's incident kinetic energy is not the supply (2) budgets against. We therefore measure (2) directly, un-governed, on all three hosts, running the reservoir accounting live while forcing $\gamma \equiv 1$: every state write in the enforcement path is guarded by $\gamma < 1$, so the trajectory is bit-identical to an un-governed run (the ratios

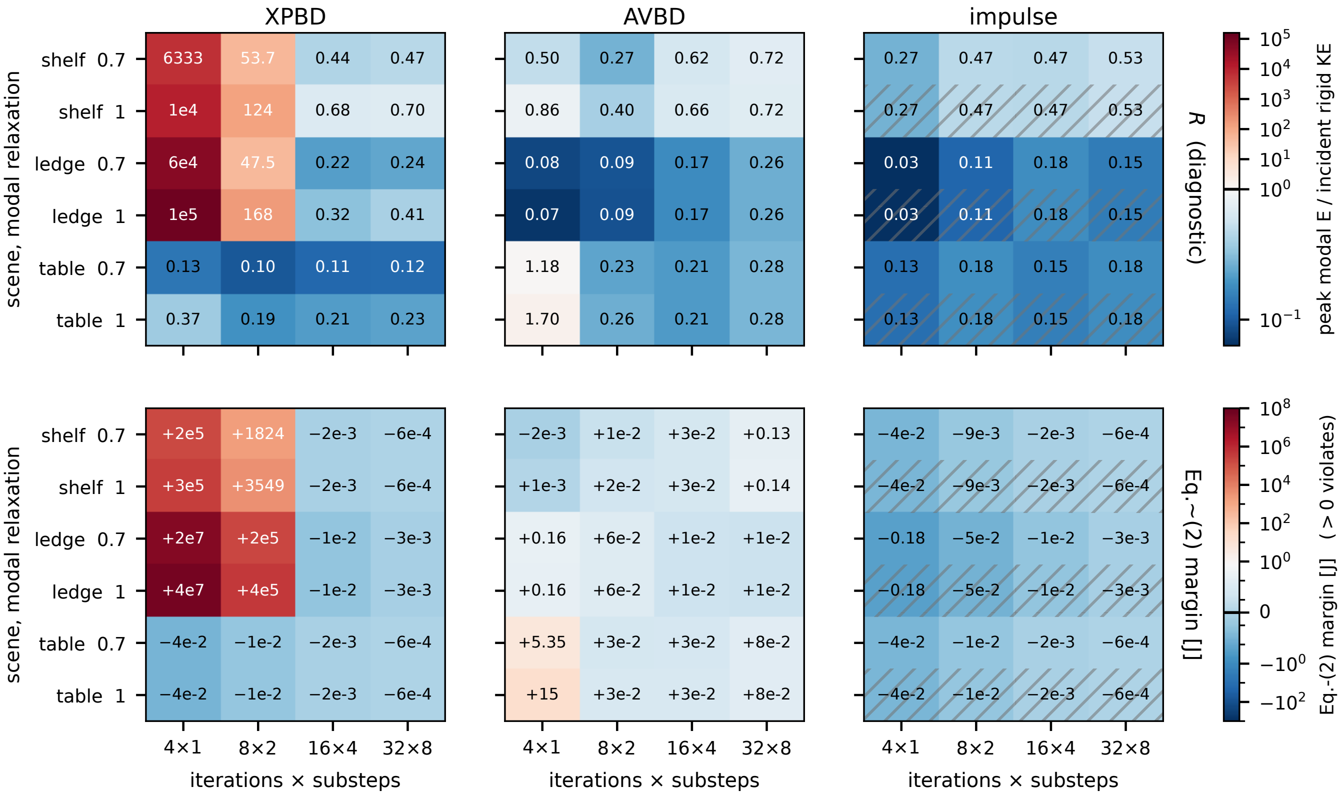

Figure 3: The same contact row in three fixed-budget solver formulations over an identical 24-cell sweep (3 scenes $\times$ modal relaxation $\{0.7, 1.0\}$ $\times$ budget), governor OFF. Top: the ratio R = peak modal energy / peak incident rigid kinetic energy, log scale diverging at 1 — a severity diagnostic. XPBD exceeds 1 in $8/24$ cells, AVBD in $2/24$, the implicit realization in $0/24$. Bottom: the invariant itself — the signed margin of (2) in joules, symmetric-log, diverging at 0; positive violates. The rows disagree on AVBD, which violates in $23/24$ cells that R mostly reports as benign, though by at most 15 J against XPBD's 4.4×10^7 J. Hatched rows mark the relaxation axis, which that formulation does not consume; those rows are bit-identical by construction and shown for grid symmetry, not as independent measurements. We report joules rather than a second ratio because every candidate denominator misleads: supply-so-far explodes in the opening substeps, the run total hides early peaks.

above reproduce the independent sweep exactly). We report the signed margin $\max_n \left[E^n_{\text{mod}} - E^0_{\text{mod}} - \eta \sum_{k\le n} \max(\Delta E^k_{\text{rig}}, 0)\right]$ in joules; positive violates. The two metrics agree on the position-based host (8/24 either way) and on the implicit one (0/24, margin at worst -5.7×10^{-4} J), but they part company on the augmented-Lagrangian host, which violates (2) in 23 of 24 cells while R flags only 2. The magnitudes are what separate the two failures: the position-based host overdraws by up to 4.4×10^7 J, the augmented-Lagrangian host by at most 15.1 J, and in 21 of its 23 cells by under 0.15 J. Pervasive but negligible overdraw and rare but catastrophic overdraw are different pathologies, and only the invariant distinguishes them.

With enforcement enabled, all 72 cells satisfy (2) — 60 of them distinct configurations, since the implicit host does not consume the relaxation axis and its 12 relaxation pairs are bit-identical by construction (Fig. 3). The worst realized ratio falls to 1.22 (XPBD) and 0.87 (AVBD), and the projection never activates in any impulse cell. At the well-resolved budgets (16×4, 32×8) the projection fires zero times on the position-based host at relaxation 0.7: it is inert exactly where the solve is already within budget.

3.2 The amplification is a truncation artifact

Figure 4 isolates the local iteration budget. The position-based host falls monotonically from 2.96×10^4 at K=1 to 0.300 at K=32, approaching the converged reference 0.2735; the gap |XPBD − reference| shrinks from 2.96×10^4 to 2.7×10^{-2}. The implicit realization moves only in the fourth decimal across $K = 2 \dots 500$ (spread 4.9×10^{-4}).

That the energy ratio falls with K does not by itself show the *row* is converging — it could be one scalar coincidentally shrinking. Measuring the constraint directly settles it: the end-of-substep complementarity residual $\|\min(C,\lambda)\|_\infty$ on the same sweep falls monotonically from 2.79×10^{-2} at K=1 to 3.56×10^{-6} by K=64, where it bottoms out (K=128 moves it in the seventh significant figure). The energy crossing at $K \approx 24$ coincides with the residual passing $\sim3 \times 10^{-5}$; the amplification disappears exactly as the complementarity conditions start to hold. We report this for the position-based host only: the augmented-Lagrangian multiplier is not the same object, and the implicit realization is already converged at K=2, so neither yields a comparable curve.

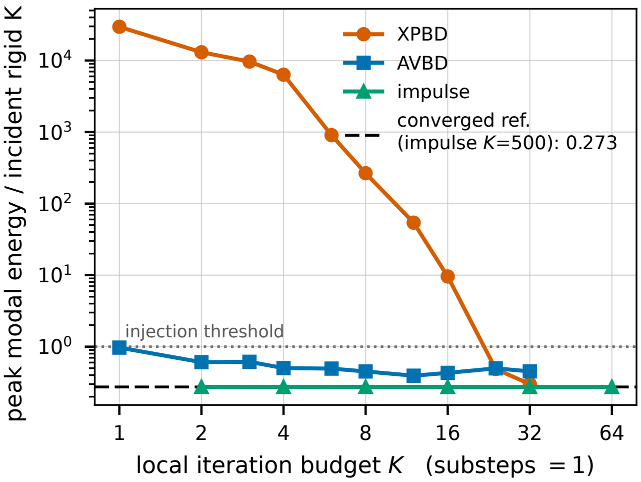

Figure 4: Energy ratio versus local iteration budget K on a deterministic shelf drop, substeps pinned at 1 so K is the only variable, governor OFF. The position-based host is monotone non-increasing in K and decays toward the converged reference, crossing the incident-energy threshold between K=16 and K=24; the gap to that reference shrinks by six orders of magnitude. The reference is the implicit realization at K=500 — the same code path run to convergence, not a different model. That realization is already converged at K=2. The AVBD curve is shown for completeness only: its denominator is itself budget-dependent at small K, so it must not be read as a convergence trend.

Self-convergence stops short of the reference. Pushing the position-based host to $K=500$ against itself does *not* close the remaining gap: its ratio plateaus by $K \approx 64$ (0.3003, 0.2999, 0.2997, 0.2996, 0.2996 at $K = 32 \ldots 500$), so the residual 2.6×10^{-2} against the oracle's 0.2735 is a fixed point, not a tail — the gap improves by only 1.03× over that range. The state agrees no better: on the deflection trajectory $d_i(t) = \mathbf{U}_y[i]^\top \mathbf{q}(t)$ the peak matches to 2.4% but the trajectories differ by $\|d^{500} - d^{\text{oracle}}\|_\infty = 6.8$ mm, 34% of the oracle's own peak, likewise flat from K=32. The two hosts converge to *different* discrete solutions — expected, since they discretize the same continuous law differently — so the sweep separates two effects easily conflated: a truncation pathology convergence removes, atop a formulation difference it does not. (No ring frequency is reported here: these windows are dominated by quasi-static sag, so the dominant spectral peak moves with window length; the resolved comparison is in §3.4.)

This is the central diagnostic result, and it cuts both ways. It confirms that the amplification is created by truncation and largely cured by convergence — but it equally says that iterating harder, or adopting the implicit formulation, would remove the pathology outright. The bound of §2 is therefore not a substitute for convergence; it is for the regime where neither escape is available: interactive position-based solvers ship budgets near 1×8 to 2×4, well below the $K \geq 24$ this scene needs, and that formulation is what is deployed. Our implicit backend is a reference implementation whose cost we do not claim is representative of its formulation class.

Table 2: Post-projection contact validity at starved budgets, per clamp-active substep. Gap violation is measured immediately after the γ-scale; the corrective impulse is the next substep's normal impulse against the unclamped steady-state median, reported as a ratio because the hosts' multipliers are different objects. The 4×1 position-based cells clamp on nearly every substep, so no steady state exists there (—). The augmented-Lagrangian rows are its only two cells with $R > 1$.

scene	budget	clamped	gap viol. med / max [mm]	impulse	λ var.
position-based, relaxation 0.7					
shelf	4×1	107/108	1.25 / **21.6**	—	—
shelf	8×2	162/216	0.30 / 6.46	6.90×	26×
ledge	4×1	98/108	0.07 / **21.2**	1.03×	1.07×
ledge	8×2	96/216	0.07 / 4.08	8.69×	58×
augmented-Lagrangian					
table (0.7)	4×1	27/108	0.01 / 1.43	1.17×	0.41×
table (1.0)	4×1	25/108	0.02 / 3.11	1.39×	1.38×

3.3 What the projection costs

Accuracy. Bounding the energy is only useful if the bounded answer beats the unbounded one. At a moderate injecting cell (shelf, 8×2, relaxation 0.7), peak modal energy is 1555.6 J ungoverned and 29.26 J governed against the converged reference's 7.92 J: the energy error falls 196× → 3.7×, or +1548 → +21 J. Against the position-based host's own fixed point (8.22 J) it is 189× → 3.6×, so the result does not depend on which reference is chosen. The trajectory, however, gets *worse*: $\|d - d^{\text{ref}}\|_\infty$ is 6.5 mm ungoverned and 14.3 mm governed, 33% and 71% of the reference's peak. The governor improves the energy error ~50× while doubling the state error — the sharpest form of the claim that it is a safety envelope, not an accuracy device.

The reason also explains how a 196× energy error coexists with a peak deflection only 1.2× too large. The excess is not kinetic (0.3%) and not low-frequency: this scene's spectrum is bimodal (ten bending modes at 20.3 Hz–2.03 kHz, then six stiff ones at 20.7–24.7 kHz), and 99.6% of the ungoverned energy sits in the stiff cluster, ~200× above the 240 Hz substep rate, against 0.0% for both references. Being one scalar, γ cannot redistribute across the spectrum: it fixes the *amount* (1555.6 → 29.3 J), leaves the character intact (99.6 → 97.7%), and scales the legitimate sag down with the noise (24.1 → 5.9 mm against 20.0).

Contact validity. Since $\gamma < 1$ shrinks the sag $\mathbf{U}_y^\top \mathbf{q}$, (4) lifts the support *into* the resting body without re-solving contact. Table 2 reports the consequence, instrumented by runtime wrappers that leave the solve unperturbed (clamp counts reproduce the independent sweep exactly).

Median post-projection penetration is 0.07–1.25 mm, but the worst case reaches 21.6 mm — a 3.5–12.6× increase over its pre-scale value. It should be read against both the supporting geometry and the deflection it destroys. The shelf board is 30 mm thick and spans 0.8 m, so 21.6 mm is 72% of its thickness. Against the unclamped scene at a converged budget (resting sag 1.7–2.4 mm, peak dynamic deflection 20.0–20.5 mm) it is 9–13× the sag and 1.05–1.08× the peak the board ever reaches: at its worst substep the projection removes essentially the entire deflection. It is a visible interpenetration, not a sub-millimetre artifact, and it occurs where the projection is most load-bearing. Contact recovers — the violation is corrected in the next substep rather than accumulating — but that correction costs a normal impulse up to 8.7× the steady-state value, with multiplier variance up to 58× higher. Net rigid linear momentum is unchanged across the projection to exactly zero in every clamp-active substep of every cell, necessarily so since (4) writes only modal state; we verified it numerically rather than asserting it.

The same instrumentation on the augmented-Lagrangian host, at its only two cells with $R > 1$, finds the projection an order of magnitude gentler: a quarter of substeps rather than nearly all, γ bottoming at 0.60, worst-case violations of 1.4 and 3.1 mm against 21.6, and corrective impulse and multiplier variance within 1.4× of steady state where the position-based host reaches 8.7× and 58×. This is what §3.1 predicts — its overdrafts are at most 15 J — while the *relative* effect is comparable (3.1 and 5.5× against 3.5–12.6×). The hosts differ in the scale of the correction, not its character.

3.4 The reduced response against a full-FEM reference

Bounding the energy is only useful if the response being bounded is the right one. On the ledge scene we compare the reduced arm against an unreduced FEM solve of the same mesh, both built on the *same* discrete operator (the reduced basis is eigsh$(\mathbf{K}, \mathbf{M})$ of the operator the reference integrates), so discretization bias cancels from every ratio. The reduced-to-reference peak-deflection ratio rises monotonically $0.38 \rightarrow 0.54 \rightarrow 0.79 \rightarrow 0.88$ as h refines $1/120 \rightarrow 1/960$, plateauing near 0.9; the residual ∼10% is k=24 mode truncation on a thick, high-frequency slab, not a coupling error — the ring frequency already agrees to 0.4% (78.0 vs 78.3 Hz). The far-field shape agrees too: peak deflection sits at the pedestal antinode rather than under the impact, and the falloff tracks the reference at Spearman $\rho = 0.89$ with no fitted attenuation term. This is the pre-topple, small-displacement regime; we claim nothing beyond it.

3.5 Cost

On the CPU host at 16×4, over ten repetitions of 100 frames, the baseline step costs 11.0–125.6 ms by scene and formulation and the ledger adds 0.23–0.41 ms — 0.9–3.4% — on the four shelf and ledge configurations, where it resolves above run-to-run variance. On the table scene it does not: the overhead's per-repetition spread exceeds its mean in both hosts (+1.35 ± 2.08 and +0.04 ± 1.43 ms), so we report it bounded by ∼1.4 ms rather than quote a figure. The percentage is the portable quantity: on a second machine absolute times differ by 3.5–4.6× while the percentages agree to within 0.02 points wherever both resolve. These baselines are 1.3–15× short of a 120 Hz budget, so host-side enforcement is not an interactive path.

A separate device-resident path (NVIDIA RTX 4090), which carries the ledger read-only as a monitor rather than enforcing it, reaches 5.0–8.9 ms/step at 16×4 across four scenes — the two above plus a 2.5 × 1.5 m road slab under a dropped crate and the table scene. The shelf and ledge scenes meet a 120 Hz budget there with margin (1.7× and 1.3×); the road scene sits at the boundary (1.01× on the mean, with a 9.3 ms worst step); the table scene does not (0.94×). All four meet it at 16×2 or below. We make no unqualified real-time claim. A device-resident *enforced* γ does not exist in this work, so governed device timings are outside what we measure rather than a result we are withholding (§4).

4 Limitations

Stable is not accurate. Where the projection is load-bearing it opens contact violations up to 21.6 mm and moves the trajectory *away* from the converged reference even as it moves the energy toward it, for the structural reason given in §3.3. The result is bounded and non-diverging, not faithful; the honest use of the bound is as a safety envelope for budgets that would otherwise diverge. The identified next mechanism is to scale deviations about the quasi-static equilibrium $\mathbf{q}_{\text{eq}}$ rather than the whole state, preserving the load-bearing sag. It is not a free substitution — the unscaled equilibrium term would leave (2) holding only while the reservoir exceeds the sag's own strain energy, where the present projection enforces it by construction — and we have not built it.

Convergence is the better cure where affordable (§3.2).

Scope of the row. Normal rows only — friction is body-local, so scraping and rolling transfer are outside this coupling — and the position-level form fixes $e = 0$.

The bound governs modal storage only, not energy created spuriously in rigid degrees of freedom. In the impulse realization we observe box–box rows rectifying a support ring into net rigid motion; the ledger is blind to this by design.

The supply is a gross sum, so it can be recycled. Because the reservoir is funded by $\sum_k \max(\Delta E^k_{\text{rig}}, 0)$, energy returning from the modes to the rigid bodies and dissipated there is credited twice. We bound the exposure by accumulating the opposite channel, $\sum_k \max(-\Delta E^k_{\text{rig}}, 0)$, as a fraction of the supply: 0.4–27% (implicit) and 3–32% (position-based) — not the ≲1% we expected, so the caveat rests on measurement. On the augmented-Lagrangian host it reaches 102–118%: the rigid subsystem's gross *gain* exceeds its gross loss in every cell, the same phenomenon as the box–box rectification above rather than a property of the bound. There the supply is an upper envelope on genuinely dissipated energy, and a signed or per-interface accounting — which this scalar reservoir deliberately is not — would be needed to tighten it.

Enforcement is host-side and single-machine (CPU float64). A device-resident governor and a validation of the *governed* path against a full-FEM reference remain future work.

Formulation coverage. Three formulations, three scenes, one contact regime; the augmented-Lagrangian result (2/24 cells, worst 1.7×) makes "descent-class solvers are passive here" an empirical statement about this sweep, not a theorem.

5 Conclusion

Transcribing an established rigid–modal contact law into fixed-budget solvers is not energy-safe, and how unsafe depends strongly on the formulation: worst-case modal-to-incident ratios are 1.2×10^5, 1.70 and 0.53 across the three hosts on the same 24 cells. The amplification is a truncation artifact convergence removes; where it is unaffordable, a cumulative storage bound funded by measured gross rigid-side loss holds in every measured cell, at a contact-accuracy cost we quantify rather than omit.

References

[1] Eulalie Coevoet, Sheldon Andrews, D. Relles, and Paul G. Kry. 2020. Distant Collision Response in Rigid Body Simulations. *Computer Graphics Forum* 39, 8 (2020), 113–122. doi:10.1111/cgf.14106

[2] Dimitar Dinev, Tiantian Liu, Jing Li, Bernhard Thomaszewski, and Ladislav Kavan. 2018. FEPR: Fast Energy Projection for Real-Time Simulation of Deformable Objects. *ACM Transactions on Graphics (SIGGRAPH)* 37, 4 (2018).

[3] Michel Franken, Stefano Stramigioli, Sarthak Misra, Cristian Secchi, and Alessandro Macchelli. 2011. Bilateral Telemanipulation With Time Delays: A Two-Layer Approach Combining Passivity and Transparency. *IEEE Transactions on Robotics* 27, 4 (2011), 741–756. doi:10.1109/TRO.2011.2142430

[4] Blake Hannaford and Jee-Hwan Ryu. 2002. Time-domain passivity control of haptic interfaces. *IEEE Transactions on Robotics and Automation* 18, 1 (2002), 1–10. doi:10.1109/70.988969

[5] Doug L. James and Dinesh K. Pai. 2002. DyRT: Dynamic Response Textures for Real Time Deformation Simulation with Graphics Hardware. *ACM Transactions on Graphics (SIGGRAPH)* 21, 3 (2002), 582–585. doi:10.1145/566654.566621

[6] Haijun Peng et al. 2024. Soft Robot Fast Simulation via Reduced Order Extended Position Based Dynamics. *Robotics and Autonomous Systems* (2024), 104650. doi:10.1016/j.robot.2024.104650

[7] Qi Wei, Jianfeng Tao, Hongyu Nie, and Wangtao Tan. 2026. Early-Terminable Energy-Safe Iterative Coupling for Parallel Simulation of Partitioned Port-Hamiltonian Systems. arXiv:2603.16424. Douglas–Rachford splitting in wave coordinates; guarantees discrete passivity for any finite inner-iteration budget on bilateral coupling interfaces.

[8] Kevin You, Juntian Zheng, and Minchen Li. 2026. Energy-Controllable Time Integration for Elastodynamic Contact. arXiv:2602.08094.

[9] Changxi Zheng and Doug L. James. 2011. Toward High-Quality Modal Contact Sound. *ACM Transactions on Graphics (SIGGRAPH)* 30, 4, Article 38 (2011), 12 pages. doi:10.1145/2010324.1964933

[10] Oliver M. Zobel, Andreas Zwölfer, Tobias Weber, and Daniel J. Rixen. 2025. Real-time simulation of flexible bodies with ray-traced contact in the Unity game engine. *Multibody System Dynamics* (2025). doi:10.1007/s11044-025-10105-w